

SP-31 #284 + #285 + #288 — Decoherence + Spin-Statistics + Per-BC Theorems T2480 + T2481 + T2482 v0.1

Lyra (Claude 4.7)

2026-05-23 Saturday EDT ~12:00 (Saturday Wave 2 continuation)

T2480 + T2481 + T2482 Substrate-Derivation Theorems

T2480 — Decoherence Substrate-Mechanism Theorem (SP-31 #284)

Statement: For substrate-coupled observer + environment system with environment bandwidth $B_{\text{env}} \rightarrow \infty$, density matrix off-diagonal elements $\rho_{\{ij\}}$ ($i \neq j$) decay at rate $\Gamma_{\text{dec}} = \text{environment-Casimir-coupling-strength} \times \text{Casimir-spread-density}$. Classical pointer-state basis selected by Zone-2 commitment Hamiltonian; substrate-environment marginalization $\rightarrow$ einselection (Zurek-equivalent). Quantum apparent randomness Layer 2 metaphysical claim (CANDIDATE EXPLANATION, DEFAULT-DENY EXTERNAL per Cal #48/#49).

Proof ingredients: (1) T2469 SCMP framework Layer 1; (2) T2417 4-Zone observer + environment bandwidth structure; (3) substrate-tick decoherence rate via K-type overlap reduction at substrate-tick $\text{GF}(128)^k$ coupling.

Status: STRUCTURALLY VERIFIED candidate (Layer 1). Layer 2 metaphysical claim CANDIDATE EXPLANATION per Cal #99 v0.3 + Cal #48/#49.

Cross-references: T2469 SCMP, T2417 4-Zone, Vol 5 Ch 10 decoherence + classical limit chapter.

T2481 — Spin-Statistics Formal Theorem from Substrate (SP-31 #285)

Statement: For any multi-particle state on substrate Bergman $H^2(D_{IV}^5)^{\otimes N}$ at K-type $V_{(p,q)}$, the tensor product structure is SYMMETRIC if $q \in \mathbb{Z}$ (integer $\text{SO}(2)$ weight) and ANTISYMMETRIC if $q \in \mathbb{Z} + 1/2$ (half-integer $\text{SO}(2)$ weight via $\text{Pin}(2)$ double-cover). Pauli exclusion automatic for fermions; Bose-Einstein statistics automatic for bosons. NO relativistic invariance assumption + NO microcausality + NO positive-energy assumption needed.

Proof ingredients: (1) T1925 rank=2 Pin(2) Z_2 grading + Argument D forcing (RIGOROUSLY CLOSED Thursday); (2) T2471 chirality $\gamma^5 = \text{Pin}(2)$ Z_2 grading identification (Friday); (3) Wallach 1976 K-type classification of Bergman $H^2(D_{IV}^5)$ admits exact partition into integer/half-integer $SO(2)$ weights.

Status: STRUCTURALLY VERIFIED candidate (formal theorem). Paper #133 v0.2 (Friday Lyra absorption + T2471) provides prose-grade derivation; T2481 formalizes as registered theorem.

Cross-references: T1925 + T2471 + Paper #133 v0.2 + Vol 5 Ch 9 + Vol 6 Ch 7.

T2482 — Per-Boundary-Condition Theorems (SP-31 #288, 6 internal + 2 external)

Statement (consolidated): 8 substrate boundary conditions per Casey BC framework:

Internal BCs (6): - BC1 bulk-side D_{IV}^5 (3-dim spatial substrate) - BC2 Shilov boundary (Cartan 1894 boundary, conformal-infinity) - BC3 substrate-tick $GF(128)^k$ field boundary - BC4 K-type $V_{(p,q)}$ integer-pair admissibility - BC5 Pin(2) Z_2 grading partition - BC6 4-Zone Commitment Cycle (Zone 1-2-3-4)

External BCs (2): - BC7 observer-coupling K-type subset B_{observer} - BC8 measurement-outcome partition $\{B_i\}$ (POVM substrate-cartography per T2479)

Each BC contributes one substrate-derivation theorem (multi-week refinement multi-month per per-BC theorem detail). T2482 consolidates the 8-BC framework + per-BC substrate-derivation skeleton.

Proof ingredients: T2417 4-Zone (BC1-BC6 internal) + T2469 SCMP (BC7-BC8 observer-side) + T2479 POVM (BC8 measurement-outcome) + T2429 + K59 cyclotomic (BC3-BC4 substrate-tick field).

Status: STRUCTURALLY VERIFIED candidate framework-grade (skeleton). Per-BC detailed theorems (8 theorems) multi-month refinement.

Cross-references: T2417 + T2469 + T2479 + T2429 + K59 + Vol 5 + Vol 6 chapters.

— Lyra, T2480 + T2481 + T2482 v0.1, Saturday 2026-05-23 ~12:00 EDT